

Lab : 3

Analysis Of Urban Heat Island

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Step 1: Convert Digital Numbers (DN) to Top of Atmosphere (TOA) Radiance

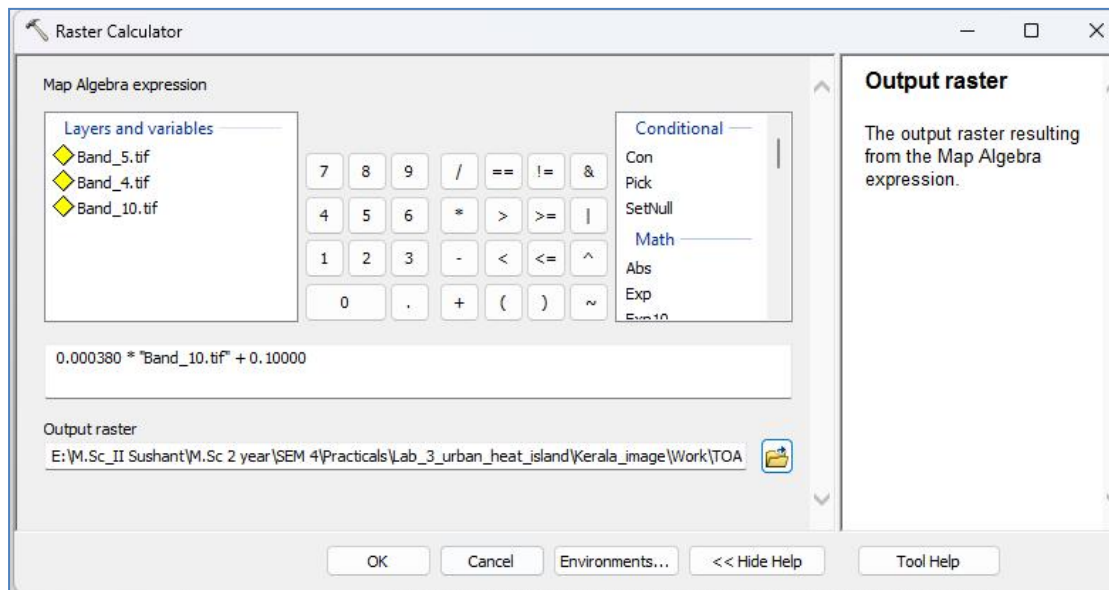
Formula :

- Use the **Raster Calculator**:

$$L_{\lambda} = (M_L \times Q_{cal}) + A_L$$

Where:

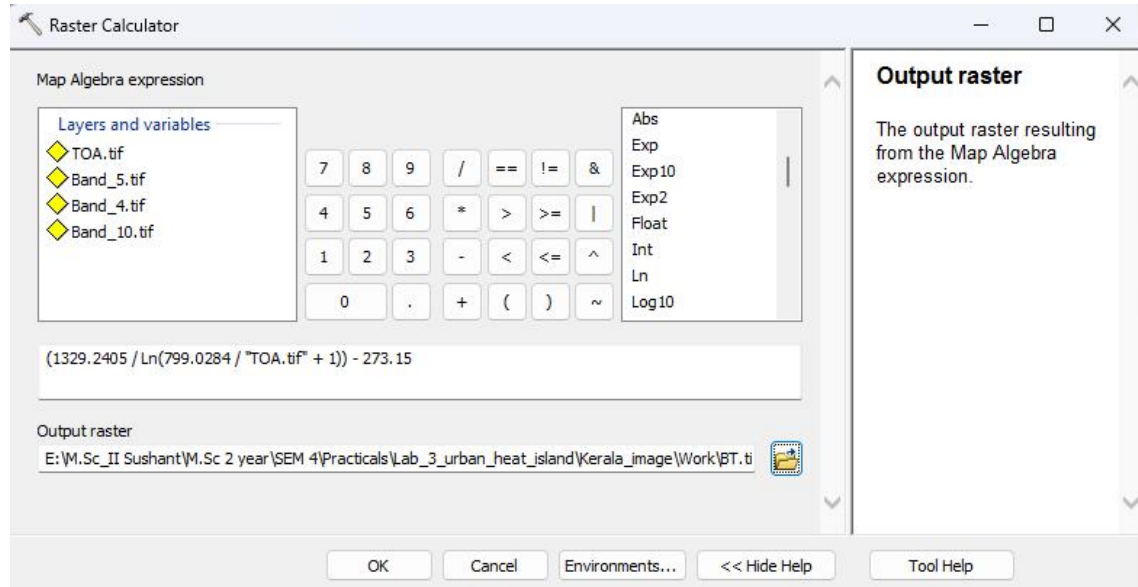
- L_{λ} = TOA Radiance ($\text{W/m}^2/\text{sr}/\mu\text{m}$)
- M_L = Radiance Multiplier (from MTL file)
- A_L = Radiance Additive (from MTL file)
- Q_{cal} = Digital Number (DN) from Band 10



Step 2: Convert TOA Radiance to Brightness Temperature (BT) in Kelvin

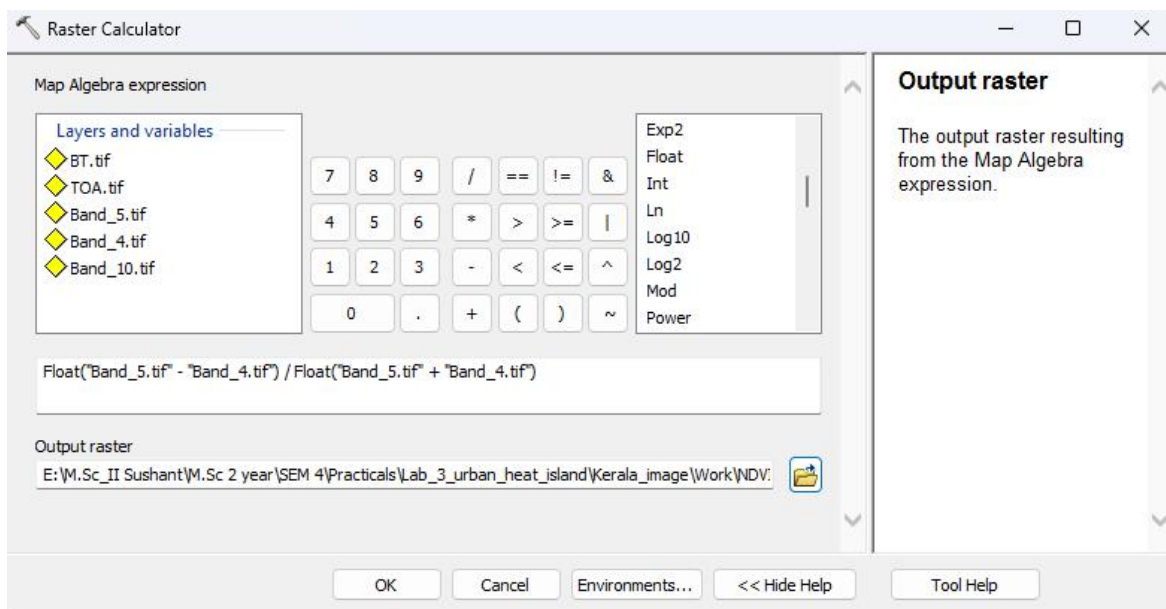
Spectral radiance data can be converted to top of atmosphere brightness temperature using thermal constant values in Meta data file

Kelvin(k) to Celsius (°C)



Step 3: Calculate NDVI

$$NDVI = \frac{(Band5 - Band4)}{(Band5 + Band4)}$$



Step 4: Calculate Land Surface Emissivity (LSE)

Land surface emissivity is the average emissivity of an element of the surface of the earth calculated from NDVI value

$$PV = ((NDVI - NDVI_{min}) / (NDVI_{max} - NDVI_{min}))^2$$

Where:

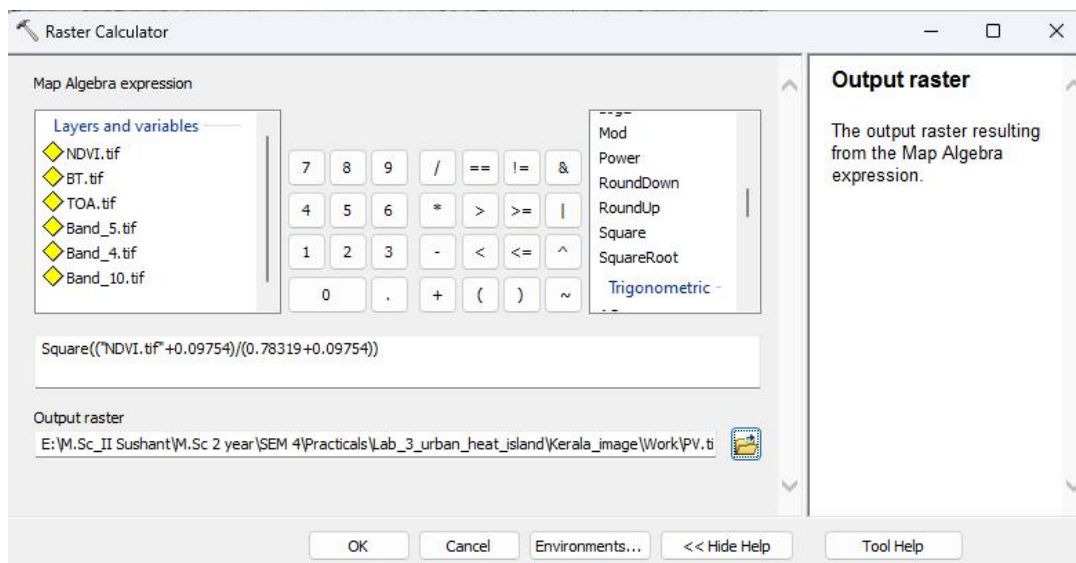
PV = Proportion of Vegetation

NDVI = DN values from NDVI Image

NDVI_{min} = Minimum DN values from NDVI Image

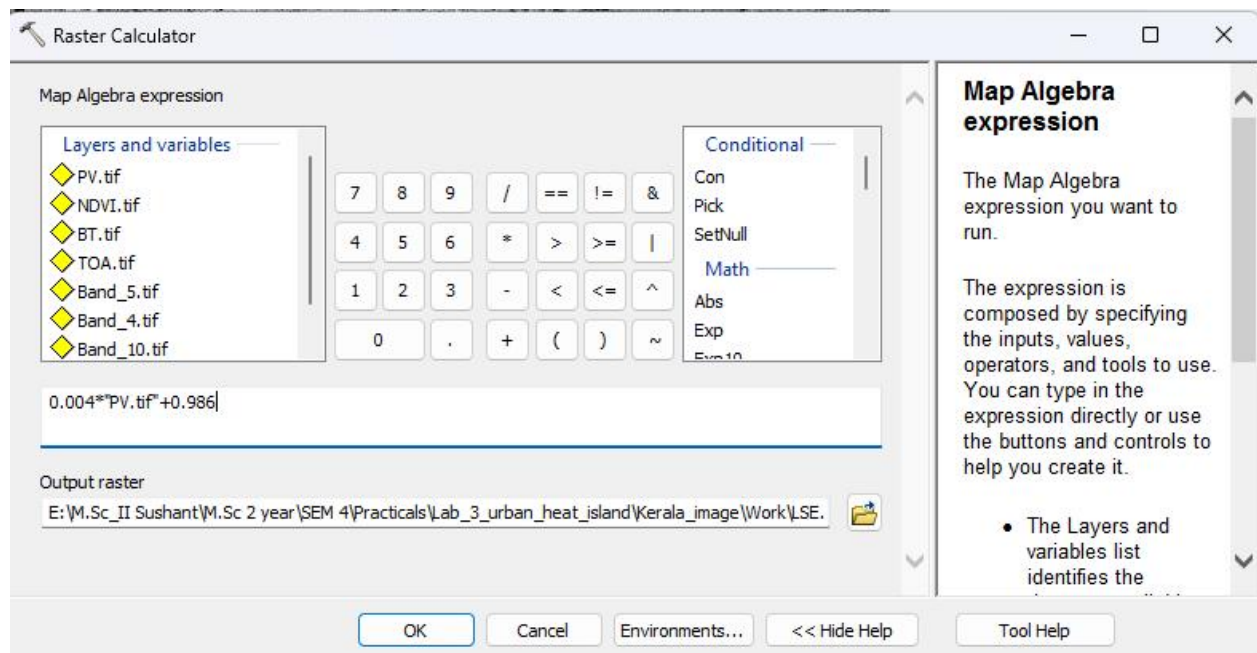
NDVI_{max} = Maximum DN values from NDVI Image

$$PV = \text{Square}((NDVI + \text{min NDVI value}) / (\text{Max NDVI value} + \text{min NDVI value}))$$



Calculate the Emissivity :

$$\text{Formula} \Rightarrow E = 0.004 * PV + 0.986$$



Step 5: Calculate Land Surface Temperature (LST)

$$LST = BT / (1 + (\lambda * BT / c2) * \ln(E))$$

Here, $c2 = 14388 \mu\text{m K}$

The Values of λ for Landsat 8: For Band 10 is 10.8 and for Band 11 is 12.0

Where

BT = Top of atmosphere brightness temperature ($^{\circ}\text{C}$)

λ = Wavelength of emitted radiance

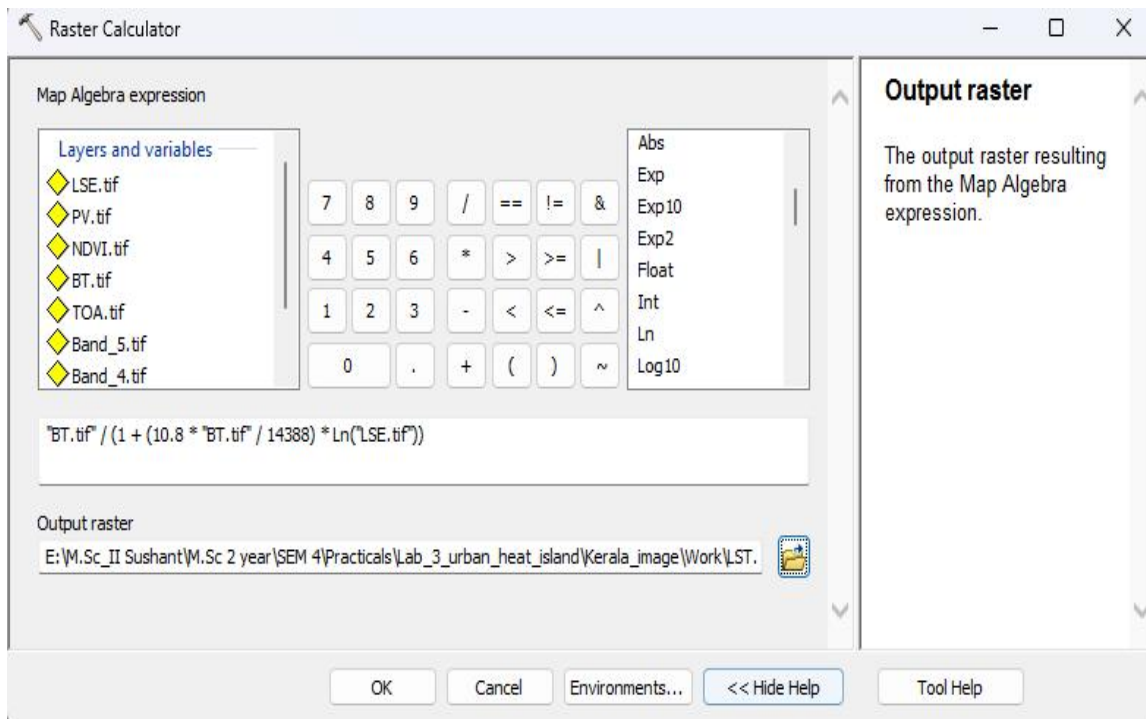
E = Land Surface Emissivity

$c2 = h * c / s = 1.4388 * 10^{-2} \text{ mK} = 14388 \text{ mK}$

h = Planck's Constant = $6.626 * 10^{-34} \text{ J s}$

s = Boltzmann constant = $1.38 * 10^{-23} \text{ JK}$

c = velocity of light = $2.998 * 10^8 \text{ m/s}$



Result (Kochi District Land Surface Temperature)

